Module 3 Reflective Journal

ITAI 1371 - Machine Learning

Neural Crew

Viviana Zamora

This lab helped me understand the three main types of machine learning more clearly. I learned that supervised learning uses labeled data with known answers, unsupervised learning looks for patterns or groups without being given the correct answers, and reinforcement learning improves through trial and error using rewards or penalties. Seeing examples of each type made the differences easier to remember and helped me understand why the choice depends on the problem being solved.

The Wine classification project also helped me understand the steps involved in a machine learning workflow. Before training a model, the data needs to be loaded, explored, checked for missing values, and divided into training and testing sets. After that, models can be trained and evaluated. I learned that accuracy provides a way to compare results, but the model and features still need to make sense for the problem.

In my notebook, Logistic Regression was the best-performing model. I think it performed better because it found the relationships between the wine features and classes more accurately than the Decision Tree for this dataset. If I continued the project, I would try different combinations of features and compare the results to see whether the accuracy improves. The most challenging part for me was understanding how to choose the best features and how to decide which model is most appropriate instead of assuming that one model will always work best.

I connected the lesson to real estate by thinking about a model that predicts home prices. That would be a supervised learning problem because the training data would include known sale prices. Useful data could include location, square footage, number of bedrooms and bathrooms, property age, and previous sale prices. This showed me that machine learning can use familiar information to make predictions that have practical value.

The most important concept I learned was how to train and evaluate different models and compare their performance. I also learned that machine learning is an iterative process: the first result is not necessarily the final result, and changes to the data, features, or model can affect performance. One question I still want to explore is how to determine which machine learning model is most likely to work well when starting with a new dataset. Overall, the lab made the machine learning process feel more understandable and less intimidating because I could see each step and its output in the notebook.

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Neural Crew

Dustin Mastrota

This lab helped me understand machine learning better than I did before. I had heard terms like supervised learning, unsupervised learning, reinforcement learning, and Logistic Regression before, but I had not really worked with them much. Actually going through the notebook and running the code made everything feel more real. Instead of only reading about machine learning, I got to see how data can be loaded, analyzed, graphed, and used to make predictions.

At the beginning of the lab, I looked at the first step where we had to import all of the essential libraries. I was honestly a little overwhelmed because there were several lines of code, and I did not know what every library was doing right away. At first, it looked like a lot to take in. I decided to slow down and go through it line by line instead of trying to understand everything at once. That helped my understanding a lot. I started to see that each library has a purpose, such as working with data, making graphs, or creating and testing the machine-learning model.

It was really cool seeing what coding can actually do when all of those libraries work together. I am still learning Python, so sometimes code can look confusing at first, especially when there are several lines in one cell. But when I ran the cells and saw the output underneath, it helped connect the code to what it was doing. I liked that Google Colab allowed me to run the code a little at a time and see the results as I worked. It made the assignment feel less like memorizing information and more like learning how the process actually works.

One part I enjoyed the most was making the graphs and analyzing the dataset. The graphs made the information easier to understand because I could actually see patterns in the Wine dataset instead of only looking at numbers in a table. Looking at the data this way helped me understand why data analysis is important before building a model. If someone does not understand the data they are working with, it would be hard to know what features may be useful or what results make sense.

I also learned more about supervised, unsupervised, and reinforcement learning. Supervised learning made the most sense to me because it uses labeled data, meaning the model has examples with known answers that it can learn from. Unsupervised learning is different because it looks for patterns or groups in data without being told the correct answers ahead of time. Reinforcement learning seems more like learning through trial and error, where an agent makes choices and receives rewards or penalties. Understanding the differences between these types of learning helped me see that machine learning is not just one thing. Different methods are used depending on the type of problem someone is trying to solve.

Towards the end of the lab, I started getting confused when I was working with the model accuracy. I was not completely sure why the accuracy could change or what I was supposed to expect when testing the model. At first, I thought that if my accuracy was lower than the original model, I must have done something wrong. After I slowed down and looked at what I changed, I realized that using different features can change the outcome of the model. Some features may give the model more useful information than others, so the accuracy can go up or down depending on what is selected. Once I understood that, the results made a lot more sense.

That was probably one of the biggest things I learned from this assignment. A machine-learning model is not just going to give the same result no matter what. The data and features that are chosen matter. Changing the input features can affect how well the model predicts the correct class. Seeing that happen in the notebook helped me understand feature selection better than just reading about it would have.

I also enjoyed learning about Logistic Regression. Before this lab, the name sounded more complicated than it does now. I am still not fully comfortable with all of the math behind it, but seeing the code run and seeing the model make predictions helped me understand the basic idea. The Logistic Regression model used information from the Wine dataset to predict the different wine classes. It was interesting to see that a computer can use patterns in data to make predictions, but that the model's results still depend on the data and features that are used.

Overall, I enjoyed this lab because it gave me more experience with coding and showed me more of what can be done with Python and machine learning. I especially liked making the graphs, analyzing the dataset, and seeing how changing features affected the model accuracy. Some parts were confusing, especially near the end when I was trying to understand the accuracy results, but working through that confusion helped me learn more. I am still new to machine learning and coding, but I feel more interested in Logistic Regression now and excited to learn more about it. I also want to get better at understanding the code, choosing useful features, and learning how to improve a model's accuracy in future assignments.

<u>Module 3: Reflective Journal</u>
<u>ITAI 1371 - Machine Learning .</u>
<u>Neural Crew .</u>
<u>Name: Funsho Orogun .</u>

This Lab allowed me to consolidate my learning. I was able to expand my understanding of ML types and Logistic Regression. I think I understand why logistic regression was a better approach for this lab, this was likely due to linear separability.

I listed concrete steps that I would take to improve the performance (like feature scaling and hyperparameter tuning). I even thought about real world applications. My field of interest is healthcare. I could see applying these concepts to a problem like predicting the likelihood of a patient developing Type 2 Diabetes.

This would be a Supervised Learning (Classification) problem. I'd need data on patient demographics, medical history, lifestyle factors, lab results (like blood glucose and HbA1c), and whether they were previously diagnosed with diabetes. The goal would be to train a model that could classify new patients as either high or low risk for developing the condition, which could help in early intervention.

To answer some of the reflection questions presented in the notebook:

<u>Which type of machine learning interests you most and why?</u>

Reinforcement Learning definitely interests me the most! The idea of an agent learning through trial and error by interacting with an environment, like teaching a robot to play chess or developing autonomous vehicles, is really fascinating. It feels very close to how humans learn in many situations, and the potential applications are incredibly exciting.

<u>What was the most challenging part of the ML workflow for you?</u>

The most challenging part for me was probably the **'Model Interpretation'** step, specifically understanding the confusion matrix and distinguishing between metrics like precision, recall, and F1-score.

It's not enough to just see an accuracy number; truly understanding why a model made certain errors and what different metrics tell you about those errors felt pretty complex initially. It requires a deeper dive than just looking at the overall accuracy.

How might you apply these concepts to a problem in your field of interest?

What questions do you have about machine learning that you'd like to explore further?

I have a few questions I'd love to explore:

1. How do you effectively handle datasets where one class is much, much larger than others (imbalanced datasets)? Like, what if only 1% of transactions are fraudulent, but you really need to catch them all?
2. What are the best strategies for selecting the most important features when you have hundreds or thousands of potential input variables?
3. How do you make sure a model doesn't just memorize the training data but can actually generalize well to completely new data, especially when you're fine-tuning a lot of parameters?

Module 3 Reflective Journal

ITAI 1371 – Machine Learning
Neural Crew
Patrick Ray

This lab helped me understand machine learning in a more complete way than I did before. Going into Module 3, I had heard the terms supervised, unsupervised, and reinforcement learning, but I did not completely understand what made them different from each other. At first, I was mostly trying to remember the definitions. After working through the examples in the lab, they started to make more sense. The easiest way for me to understand supervised learning was to think of it as learning with an answer key because the data already has labels. Unsupervised learning is different because the computer has to look through the data and find patterns or groups without being given the answers. Reinforcement learning also became easier to understand when I thought about it as learning through trial and error and receiving rewards or penalties based on the decisions that are made.

Another thing I learned from this lab was that there are several steps involved in building a machine learning model. Before doing this assignment, I did not realize how much work happens before a model actually makes a prediction. Looking at the Wine dataset helped me see why you first need to understand the data and make sure it is usable. The dataset had 178 samples, 13 features, three different classes, and no missing values. Seeing this information made me realize that checking and understanding the data is an important part of the process. If the data has problems or if you do not understand what you are working with, it could affect everything that comes after it.

The part of the lab that helped me understand this the most was experimenting with different features in the Wine classification model. I selected malic acid, flavanoids, and hue as my three features. My model reached about 83.3% accuracy, while the original model had about 88.9% accuracy. At first, I thought getting a lower accuracy meant that I had done something wrong. After looking at the results, I realized that this was actually showing me something important about machine learning. The features you choose can have an effect on how well a model performs. Just because data is available does not mean that every combination of features will give you the same results. This made feature selection feel more important to me than it did when I was just reading about it.

I also learned more about Logistic Regression during this lab. Before this assignment, the name sounded complicated because I am still fairly new to machine learning models. Seeing it actually work with the Wine dataset made it less confusing. The model was able to use information from the wine data to predict the different wine classes. I am still learning how the math behind these models works, but being able to run the code and see the accuracy made the idea easier to understand. It also made me interested in seeing what would happen if I tried other combinations of features or completely different models.

One of the more challenging parts for me was understanding how all the different steps connect together. There is a lot happening between loading a dataset and getting a final prediction. I had to slow down and look at the outputs instead of just running the cells and moving to the next one. I am also still getting comfortable with Google Colab and Python code. When I first see a larger block of code, it can look confusing, but I am starting to get better at looking at smaller parts of it and understanding what they are doing. Running the code and seeing the results right underneath it has helped me learn because I can connect the code to an actual result.

Overall, this lab gave me a better understanding of what a machine learning workflow actually looks like. I still have a lot to learn, especially when it comes to choosing the best features and understanding how different models work. However, I feel more comfortable with these concepts than I did before completing the lab. The biggest thing I took away from this assignment is that building a machine learning model is not just about running code. The data you use, the features you select, the type of learning you choose, and how you evaluate the results all matter. Going forward, I would like to learn more about how to improve a model's accuracy and how someone decides which model and features are best for a real-world problem.